Data Cleaning and Quality Assessment
Homework 4 - Anomaly Detection
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Question 1 - A Famous Experiment

Results and interpretation

I used the IQR method to identify unusual measurements in the Michelson-Morley dataset. I applied it separately within each experiment and across the entire dataset.

The two approaches can produce different results because they use different reference distributions. The within-experiment method evaluates each measurement against runs from the same experiment. The global method evaluates all observations together, without accounting for experimental grouping.

I compared the experiment to the means before and after removing within-experiment outliers. The tables and plots show how outlier removal affects each meaning and the range of means across experiments.

To determine whether the differences between experiment means increased or decreased, compare the range printed by the notebook:

- If the range after removal is larger, the differences increase.
- If the range after removal is smaller, the differences decreased.
- If the ranges are similar, the overall differences changed very little.

Conclusion: Outlier detection should consider the structure of the data. A global IQR threshold may overlook unusual observations within a particular experiment or flag observations that are reasonable for their experimental group. Outliers should be investigated before being removed because they may reflect genuine experimental variation rather than data-entry errors.

Question 2 - Misspelled Words

Results and interpretation

I analyzed the name frequencies and used string similarity to identify rare spellings that resemble more common names. Each candidate includes its observed spelling, frequency, suggested match, and similarity score.

Frequency helps identify unusual spellings, while string similarity helps prioritize those that may be typing errors. However, these measures cannot determine with certainty whether a name is incorrect.

A rare name may be legitimate, and two similar spellings may refer to different names. The most frequent similar spelling is not necessarily the correct replacement. The suggested corrections therefore require manual validation.

Conclusion: Combining frequency analysis with string similarity provides a practical way to identify possible misspellings. The original data should be preserved, and suggested corrections should be stored separately until verified. This approach reduces the risk of changing valid names incorrectly and makes the cleaning process transparent and reversible.